

**Long-term monitoring in Barkley Sound: A temporal analysis of intertidal biodiversity
on Wizard Islet, British Columbia from 1997 to 2017**

by

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Abstract

The evaluation and understanding of biodiversity is a necessary but challenging requirement for advancing conservation and management efforts. Baselines and long-term monitoring programs are critical in recognizing the magnitude of environmental changes over time as well as understanding the resiliency of ecosystems. The marine rocky intertidal zones of British Columbia are inhabited by diverse, highly productive communities that are regularly exposed to both natural and anthropogenic stressors. These stressors may be exacerbated by climate change and other perturbations. Using a baseline established in 1997, as well as subsequent data collected in 2001, 2007, and 2017, we observed changes in biodiversity at the intertidal zone on Wizard Islet, Barkley Sound, British Columbia. In 2017, we found that a distinct change in species richness occurred, as well as a shift in dominant phyla assemblages through the years analyzed. We speculate that these changes will become more prevalent over time and suggest that long-term monitoring continues in order to compare biodiversity trends under future climate change scenarios.

Keywords

Climate change, Intertidal zone, Long-term monitoring, Marine ecology

Introduction

There is a shortage of preliminary data available for comparison in modern ecological work, particularly in the marine realm. Describing and assessing marine biodiversity is necessary for understanding the dynamic nature of these ecosystems. The intertidal zone represents a transition between marine and terrestrial ecosystems, thereby providing unique habitats. A wide variety of taxa inhabit the rocky intertidal zone of coastal British Columbia, including algae, bryozoans, cnidarians, crustaceans, echinoderms, and molluscs. These organisms are subject to natural stressors and rapid changes in environmental conditions, such as wave and current forces (Kitzes and Denny, 2004), desiccation (Foster, 1971), abrupt fluctuations in salinity (Burnett, 1997; Davenport and Macalister, 1996; Li and Brawley, 2004), oxygen availability (Dahlhoff and Menge, 1996), and nutrient levels (Mieszkowska, 2016). These conditions push intertidal organisms to their physiological tolerance limits (Brown, 1984; Brown et al., 1996; Lewis, 1986; Stillman and Somero, 2000; Stillman, 2002), and these species are distributed throughout the intertidal accordingly (Harley et al., 2006).

In addition to the highly stressful and constantly fluctuating environment in which intertidal organisms live, the rocky intertidal shores of Barkley Sound, British Columbia

are littered with marine debris (pers. obs. 2017), and are also subject to chronic oil spills and high shipping traffic (Serra-Sogas et al., 2008). Climate-related parameters are also known to be a factor contributing to the spatial arrangement and distribution of intertidal communities. Sea surface temperature (SST) is often correlated with intertidal community patterns (Harley et al., 2006), and elevated air temperatures can increase thermal or desiccation stress, having strong effects on intertidal organisms (Barry et al., 1995; Stillman, 2002). As a result, some of the primary responses to climate change can be found in the rocky intertidal ecosystem, which is one of the most sensitive natural systems to climate (Bertness et al., 1996; Helmuth et al., 2006; Helmuth et al., 2011). This heightened sensitivity makes it an ideal model system to study the effects of climate change (Barry et al. 1995; Mieszkowska, 2016).

Understanding spatial and temporal changes in biodiversity is a key component of effective management, particularly when the eventual effects of climate change are not yet clear (Harrison et al., 2014). In order to detect changes in natural communities and explicitly relate them to anthropogenic effects or environmental stressors, proper biological baseline data are critical (Cruz-Motta et al., 2010). Consequently, a systematic approach to data collection and monitoring is needed in order to assess long-term temporal changes in biodiversity and to reduce erroneous significance. Moreover, measuring and recording biodiversity reduces the occurrence or persistence of shifting baseline syndrome (Pauly, 1995). This syndrome is a deterrent to effective management, in that it creates a false sense of ecological security (Hanazaki et al., 2013). Though it may seem like current biodiversity is rich and robust, it may pale in comparison to biodiversity seen generations ago. Therefore, the measuring and recording of baseline data is paramount in understanding the actual health of an ecosystem.

Here we reassess intertidal biodiversity at Wizard Islet in Barkley Sound, British Columbia. Using a long-term intertidal monitoring program established in 1997, we describe the distribution of intertidal invertebrates and macroalgae, and the abundance

of sessile invertebrates and macroalgae. We evaluate temporal changes in both species richness and percent cover of assemblages across four different years (1997, 2001, 2007, 2017) along the rocky intertidal zone of the islet. Trends between percent cover of organisms and the climatic parameter of mean SST were also analyzed in an attempt to understand the mechanism by which biodiversity changes occurred.

Materials and Methods

Site description

Wizard Islet (48.8581° N, 125.1598° W) is a small (0.2 km²) island roughly 3 km northwest of Bamfield in Barkley Sound, British Columbia. Home to the Nuu-chah-nulth First Nations, portions of Barkley Sound are included in the Pacific Rim National Park Reserve. It is a relatively low impact site as it is mainly used for research through the Bamfield Marine Sciences Centre. Located in Trevor Channel, Wizard Islet experiences average monthly air temperatures ranging from 5.6 °C to 14.1 °C and an annual average rainfall of 2735 mm (Weather Network, 2017). The islet can experience waves up to 8.45 m in height (Fisheries and Oceans Canada, 2017). The site sampled was a relatively homogeneous fixed-rocky intertidal zone located on the exposed, northwest edge of the islet.

Biological sampling

Data were previously collected at both an exposed and sheltered site at Wizard Islet in August of 1997, 2001, and 2007. In 2017, data were collected exclusively at the exposed site due to a lack of feasibility. At the exposed site, 15 transects were randomly selected from a group of 50 permanent transects. These permanent transects were previously established by students in 1997 as a baseline for intertidal biodiversity surveys, and are marked by steel bolts at 1 m intervals at a tidal height of 3.7 m.

Tidal height along each transect was determined using a laser level and stadia rod. A top-down approach was used to measure tidal height, in which one individual would set up the laser on top of the steel bolt at 3.7 m, while another individual held the stadia rod while moving toward the water. The height of the laser was measured at each of the

randomly selected steel bolts. This height was then added to 0.2 m to determine the value the laser point must reach on the stadia rod in order to identify the position of tidal height 3.5 m along the transect. 0.5 m was continually added from there to determine the location of each consecutive tidal height along each transect.

Tidal heights were used instead of intertidal zonation (i.e. “low”, “mid”, and “high” intertidal) to ensure the repeatability of sampling over time, as these zones may often be determined arbitrarily. Along each of our randomly selected transects, a 25 cm² quadrat was positioned with its bottom right corner on the tidal height marker, while facing away from the water. Nondestructive methods were used to survey biodiversity at the following tidal heights: 0.2, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, and 3.5 m. Presence or absence of species were recorded to determine species richness per quadrat, while abundance of colonial fauna and macroalgae were estimated by percent cover. Algae were often found growing as an overstory, and in this case the percent cover of the overstory was recorded and then shifted aside before searching the understory. Most species identification was conducted in the field on living organisms, although occasional troublesome specimens were photographed for reference, and later identified by personnel more familiar with intertidal species. All organisms were identified to the lowest taxon possible, which was most often at the species level.

Data analyses

Selecting transects at random allowed for a randomized sampling design, which in turn meets assumptions for many statistical tests. It is suggested that accounting for potential statistical tests leads to a more accurate and useful data set (Smart and Granger, 1974).

In order to test for statistically significant variation of mean richness between the four years, an ANOVA was used. A QQ-plot was first created to visually see how normally distributed the data was. Shapiro-Wilks was then used to statistically test for normality within the data, while a Bartlett test was used to ensure homoscedasticity of variance. As the original data set was not normally distributed, it was square transformed to meet the assumptions of ANOVA. Following the ANOVA, an ANCOVA test was conducted in

order to incorporate the effect of variation in the data caused by tidal heights. In this statistical analysis, tidal height exists as a nuisance variable, which has a dramatic effect on species richness. This in turn will affect whether or not statistical significance is identified when using ANOVA and by decomposing the effect of tidal height, the true variance between species richness and year may be analyzed. ANOVA and ANCOVA statistics as well as the two assumption tests performed were executed in RStudio version 1.0.153.

A Shannon-Wiener index was conducted to evaluate the diversity of percent cover found in 1997, 2007, and 2017. The biodiversity index was not used on the 2001 data as percent cover was not accounted for that year. Percent cover of each species was summed across all tidal heights and the proportion of percent cover for each species was determined. The proportion of each species was then multiplied by its natural log and the values were summed to establish the biodiversity index of each year,

$$H' = - \sum_{i=1}^k p_i \log_e p_i$$

whereas p_i is the proportion of percent cover, and $\log_e$ is the natural log of p_i . Graphs were created in Microsoft Excel version 15.36.

Results

Species richness in 2007 was found to be the highest of all four years, with 69 species identified in total (Fig. 1). Close behind were 2017 and 2001 with 67 and 57 species, respectively (Fig. 1). Species richness was the lowest in 1997 as only 38 species were identified (Fig. 1). Through the years, there appears to be an increase in total species richness until a slight decrease in 2017 (Fig. 1).

Mean species richness appears to be lower at the extreme high and low tidal heights while remaining relatively high in the mid-intertidal, and this trend is consistent through all years (Fig. 2). Mean species richness per year, per tidal height was highest between

tidal heights 1.0 m to 2.0 m across all four years (Fig. 2). The highest richness was observed in 2001 at 2.0 m (12.86 ± 0.64 SE; Fig. 2), while the lowest mean richness was found in 2017 at tidal height 3.5 m (0.67 ± 0.16 SE; Fig. 2). The year 2017 consistently had the lowest mean richness compared to the other three years across all tidal heights (Fig. 2). The lowest mean species richness across all four years was found at the tidal height of 3.5 m (Fig. 2).

An ANOVA was conducted on mean species richness per tidal height, per year, to determine if there was a statistically significant change in richness between the four years. The results of this test suggest that no significant difference in richness exists between the years (Table 1). Following the ANOVA, an ANCOVA was conducted as it was speculated that tidal height might act as a nuisance variable, and have a considerable effect on species richness. Although the ANCOVA revealed that the mean species richness between years was also not significant, it appears to be approaching statistical significance (Table 1). This suggests that tidal height did have a pronounced effect on mean species richness as a covariate and also that there was more variance between years than the ANOVA suggested.

Percent cover was determined for eight different phyla over the years 1997, 2007, and 2017. No percent cover data is available for 2001. Percent cover of Porifera was highest across all three years at tidal heights 0.2 m to 1.5 m (Fig. 3A). The highest occurrence of Porifera was in 2007 at a tidal height of 1.0 m (percent cover = $12.00\% \pm 7.05\%$ SE; Fig. 3A). This was double that of the second highest percent cover of Porifera, which was found at 0.2 m in 2017 (percent cover = $6.00\% \pm 1.83\%$ SE; Fig. 3A). Few Porifera species were identified in all three years between tidal heights 2.0 m to 3.5 m (Fig. 3A). Of the three years, 1997 had the lowest percent cover of Porifera (Fig. 3A).

Percent cover of Cnidaria was highest for all three years at tidal heights 1.5 m and 2.0 m, with very little percent cover at heights 0.2 m, 0.5 m and at 3.5 m (Fig. 3B). The highest occurrence of Cnidaria species was found at 1.5 m in 2017 (percent cover = $9.66\% \pm 2.63\%$ SE; Fig. 3B) while 2007 had the least percent cover of Cnidaria across most tidal heights (Fig. 3B).

Percent cover of Bivalvia was concentrated between tidal heights 2.0 m to 3.0 m, with very little cover found at the lowest and highest tidal heights (Fig. 3C). The highest occurrence of Bivalvia was found in 1997, at tidal height 2.5 m (percent cover = $36.8\% \pm 9.72\%$ SE; Fig. 3C) while 2007 had the lowest overall percent cover of bivalves between the three the years (Fig. 3C).

Percent cover of Arthropoda was dominant at tidal height 3.0 m between all three years, with the highest occurrence in 1997 (percent cover = $78.06\% \pm 4.55\%$ SE; Fig. 3D).

Percent covers were relatively low for all years between tidal heights 0.2 m and 1.0 m (Fig. 3D). All years appear to have steadily increased in percent cover as tidal height increased, peaking at tidal height 3.0 m before decreasing once more at 3.5 m (Fig. 3D).

Percent cover of Bryozoa was concentrated at the lower tidal heights, from 0.2 m to 1.0 m, and declined as tidal height increased (Fig. 3E). The highest percent cover of bryozoans was found in 1997 at 0.2 m (percent cover = $65.06\% \pm 7.40\%$ SE; Fig. 3E).

Few bryozoan species were present in 2017 when compared to the two other years, with the highest percent cover located at 0.2 m accounting for only one-eighth of the highest percent cover found in 1997 (percent cover = $7.90\% \pm 2.19\%$ SE; Fig. 3E).

Percent cover of green algae followed a bell-shaped curve, peaking between 1.5 m and 2.5 m (Fig. 3F). The highest percent covers were in 1997 at 2.0 m (percent cover = $25.80\% \pm 4.57\%$ SE; Fig. 3F) and at 1.5 m (percent cover = $21.33\% \pm 4.47\%$ SE; Fig. 3F). Percent cover in other years across all tidal heights was lower; for example, the next peak was in 2017, representing half of that of 1997 (percent cover = $12.40\% \pm 2.25\%$ SE; Fig. 3F).

Percent cover of Rhodophyta was consistently highest in 1997 between tidal heights 0.2 m and 2.0 m with the highest coverage being at 0.5 m (percent cover = $119.87\% \pm 10.88\%$ SE; Fig. 3G). Percent cover here exceeds 100% as multiple layers of Rhodophyta were considered (Fig. 3G). Very little Rhodophyta was found between tidal heights 2.5 m to 3.5 m for all three years represented (Fig. 3G).

Percent cover of Phaeophyceae was highest in 2007 across most tidal heights, with the highest percent cover occurring at 1.0 m (percent cover = $75.07\% \pm 9.16\%$ SE; Fig. 3H). The percent cover of Phaeophyceae was high across a wide range of tidal heights and was not concentrated around a particular height (Fig. 3H), however it was highest in both 1997 and 2007 at 1.0 m (Fig. 3H). Percent cover of brown algae in 2017 was relatively lower than other years, yet remained fairly consistent across all tidal heights before decreasing to 0.00% in the highest tidal height, 3.5 m (Fig. 3H). It is evident that phyla assemblages have shifted through each of the four years (Fig. 3).

A scatterplot was used in order to compare total overall species richness per year to the mean SST in the Barkley Sound region (Fig. 4). This plot demonstrates a slight negative trend between SST and species richness, suggesting that increased SST leads to a decrease in species richness. However, this trend is weak due to only four data points (Fig. 4). SST data for August 2017 was not available at the time of this analysis. Data was collected from Amphitrite Lighthouse (Fisheries and Oceans Canada, 2017).

A Shannon-Wiener Biodiversity Index calculation revealed that 1997 saw the highest richness of percent cover ($H' = 1.29$) when compared to the other years. 2007 had the second highest biodiversity score ($H' = 1.17$) and 2017 had the lowest biodiversity ($H' = 1.21$).

Discussion

Using a biodiversity baseline established in 1997 at Wizard Islet, we found that overall intertidal species richness increased from 1997 to 2007, before declining from 2007 to 2017. Intertidal community assemblages and dominant phyla composition also varied across all four years evaluated. Bryozoa for instance were extremely dominant in the lower intertidal regions during 1997 but appear to have decreased dramatically in the years since. Another study has found a dramatic decrease in starfish density in the Barkley Sound area since the late 1990's (LaLiberte and Morgan, in press). We speculate that this decrease in a keystone species may have resulted in a top down shift in population structure, in this case, an increase in bivalve population sizes, which is reflected in our data. As conditions become more favourable for certain organisms, it is

expected we will see further shifts as species will either adapt or disappear from the gene pool altogether.

Differences in intertidal species richness across tidal heights demonstrated that spatial distribution patterns of intertidal biota were not homogenous over spatial nor temporal scales. Irrespective of temporal change, we found that mean species richness peaks in the mid-intertidal (1-2 m), and is lowest in the low- (0.2-0.5 m) and high- (2.5-3.5 m) intertidal. This spatial zonation of organisms has been attributed to occupation of different niches, as environmental heterogeneity allows species with differing traits to persist, particularly when different species perform best at different time or places (Boese et al., 2003; Moschino and Da Ros, 2009; Whalen and Aquilino, 2006).

The stressors associated with diurnal tidal cycles, coupled with seasonal variations in both air and sea temperatures, mean that intertidal organisms are subject to extreme thermal fluctuations, and consequently many are already living at the edge of their physiological tolerance limits (Lewis, 1986; Brown et al., 1984; Brown et al. 1996; Stillman and Somero, 1996; Stillman, 2002; Helmuth et al., 2006). Consequently, intertidal organisms may be some of the earliest to respond to climate change (Helmuth et al., 2006; Mieszkowska, 2016). Tracking temporal changes in marine systems, particularly intertidal systems, then becomes extremely crucial for strengthening our ecological understanding, and for increasing our ability to separate natural variation from anthropogenic impacts (Irvine and Shelley, 2012).

This interaction between environmental variability and niche differences explains our bell-shaped curve in species richness across tidal heights; organisms in the high-intertidal are more frequently exposed to radiation and desiccation stress (Foster, 1971), while organisms in the low-intertidal are more frequently exposed to current and wave forces and elevated predation pressure (Lubchenco and Menge, 1978; Kitzes and Denny, 2004). There are fewer organisms that are able to successfully exploit these niches and withstand these stressors, which explains the distribution of species richness across various tidal heights (Boese et al., 2003; Moschino and Da Ros, 2009; Whalen and Aquilino, 2006).

When evaluating the total number of species identified in each year, 2017 contains the second highest number of species identified. This may suggest an increase in the different types of organisms habituating Wizard Islet, particularly when comparing to the number of species identified in 1997, which is nearly half of the number of species identified in 2007. While this may suggest a shift in the spatial distribution of niche availability, it may also be a consequence of inconsistent sampling efforts of previous surveyors. For example, surveyors in 1997 did not identify any motile invertebrates, such as annelids, crustaceans, echinoderms, and gastropods. This may be attributed to an absence of these invertebrates, however this is improbable because these invertebrates were found from 2001 onwards, and there is no literature to suggest that any die-offs of these groups occurred in our study region in the late 1990s. However, a die-off in sea stars in 1997 was recorded in southern California (Eckert et al., 2000). We suggest that the lack of these invertebrates surveyed in 1997 was an oversight by the surveyors, as we did not find any obvious or apparent loss of guilds or genres.

Although no motile invertebrates were recorded in 1997, mean species richness across tidal heights in 2017 is lower than that of 1997. This may indicate that although an overall wider variety of organisms were identified in our survey, surveyors in 1997 may have identified a greater variety of organisms per quadrat. Ultimately, this intertidal shore of Wizard Islet may have been able to support a wider variety of organisms in 1997 than 2017, as reflected in the higher Shannon-Wiener Biodiversity Index ($H' = 1.29$) for this year. However, it is important to note that because the Shannon-Wiener value and species richness are different metrics of biodiversity, it is unlikely that they will show the same trends. For example, 2007 had the lowest Shannon-Wiener value ($H' = 1.17$), but had higher overall species richness than 2017.

2017 had the lowest mean species richness across tidal heights, which is approaching statistical significance. This warrants concern when considering how intertidal community composition is expected to change under future climate change scenarios. For example, the sessile nature of colonial intertidal ectotherms prevents escape from changing environmental conditions (Mieszkowska, 2016). Moreover, macroalgae and invertebrates inhabit low trophic levels and thus would be expected to respond quicker

to alterations in environmental regimes than species at higher trophic levels, often displaying the first response in a trophic cascade (Hilmuth et al., 2011).

Ocean acidification is also predicted to influence intertidal community composition. The abundance of calcifying rocky shore invertebrates, such as limpets and barnacles, is expected to decline in the second half of this century (Helmuth et al., 2006; Mieszkowska 2016). The effects of changing seawater pH is concerning, as is shown by experiments on algae. These effects can include regime shifts from kelp forests to turfing algal communities, with associated loss of biodiversity (Monaco et al., 2014). However, accurately predicting the extent and rate of future spatial shifts in intertidal assemblages requires a better understanding of the biological mechanisms driving these changes (Fortin et al., 2005).

The effects of climate change on rocky intertidal communities are expected to be widespread, though species richness was poorly correlated with SSTs in our study. This may be attributed to small sample size, in that only four years of biodiversity data were available for analysis. Additionally, given that SST data for August 2017 (the month when our surveys were conducted) was not available at the time of this analysis, and SST data for July 2017 was used instead. This warrents concern when comparing SSTs between the years for which data is available. Moreover, the nearest mean SST data is available from Amphitrite Lighthouse, 28.8 km away from Wizard Islet. A strong correlation may be seen if these biodiversity surveys continue into the future, and if data pertinent to Wizard Islet becomes available.

Although this correlative analysis did not establish a cause-consequence relationship between SST and species richness, future research should consider identifying correlated drivers of biodiversity, which can give insight on which climatic parameters may be most influential. Definitive cause-effect relationships between environmental drivers, whether natural or anthropogenic, and rocky intertidal assemblages are further complicated by the complexity of spatial and temporal variation in which natural fluctuation of these assemblages occur (Harley et al., 2006; Moore et al., 2007; Cruz-Motta et al., 2010).

Despite the caveats of the data in this study, we have shown the importance of robust long-term monitoring and maintaining databases of ecological information to enhance understanding of the structure and functioning of rocky shore assemblages. Long-term monitoring programs can be used to determine spatial and temporal variation in rocky marine intertidal ecosystems, thereby facilitating our understanding of the level of resilience and resistance of these ecosystems following disturbances. Implementing long-term monitoring can also reduce the prevalence of the shifting baseline syndrome. Although this phenomenon is most prevalent in the context of fisheries management (Pauly, 1995), regular assessment of intertidal biodiversity can reduce the likelihood of unnoticed reductions in populations, extirpation, or extinction (Carlton et al., 1991).

Sites selected for long-term biodiversity monitoring need to be spatially representative in order to determine the spatial scale of the disturbance or perturbation. In Barkley Sound, long-term monitoring sites should include various substrate types beyond the fixed-rock substrate represented in our survey. These include moveable rock (i.e. cobble) beaches, sand, and mud, and should also include different aspects and varying degrees of wind and wave exposure. We advise that the Bamfield Marine Sciences Centre take on a large role in the establishment of more long-term monitoring programs throughout the Barkley Sound area.

In order to develop conservation strategies that will work in the long-term, it is necessary to recognize that change in ecological systems will occur regardless of human intervention and, understand the problems of assessing and interpreting this change. We are currently experiencing a global biodiversity crisis, and we expect to see further alterations in rocky intertidal communities and other ecosystems. It is our hope that BMSC students continue this monitoring program and are able to experience the same wealth of diversity as we did.

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Tables and Figures

Table 1

Summary of ANOVA and ANCOVA results analyzing for significance of mean species richness between 1997, 2001, 2007, and 2017, at Wizard Islet. Each mean value is the result of 15 replicates. ANOVA and ANCOVA were run in RStudio version 1.0.153.

Statistical test	df	f-value	p-value
ANOVA	3	1.91	0.15 (> 0.05)
ANCOVA	3	2.72	0.06 (> 0.05)

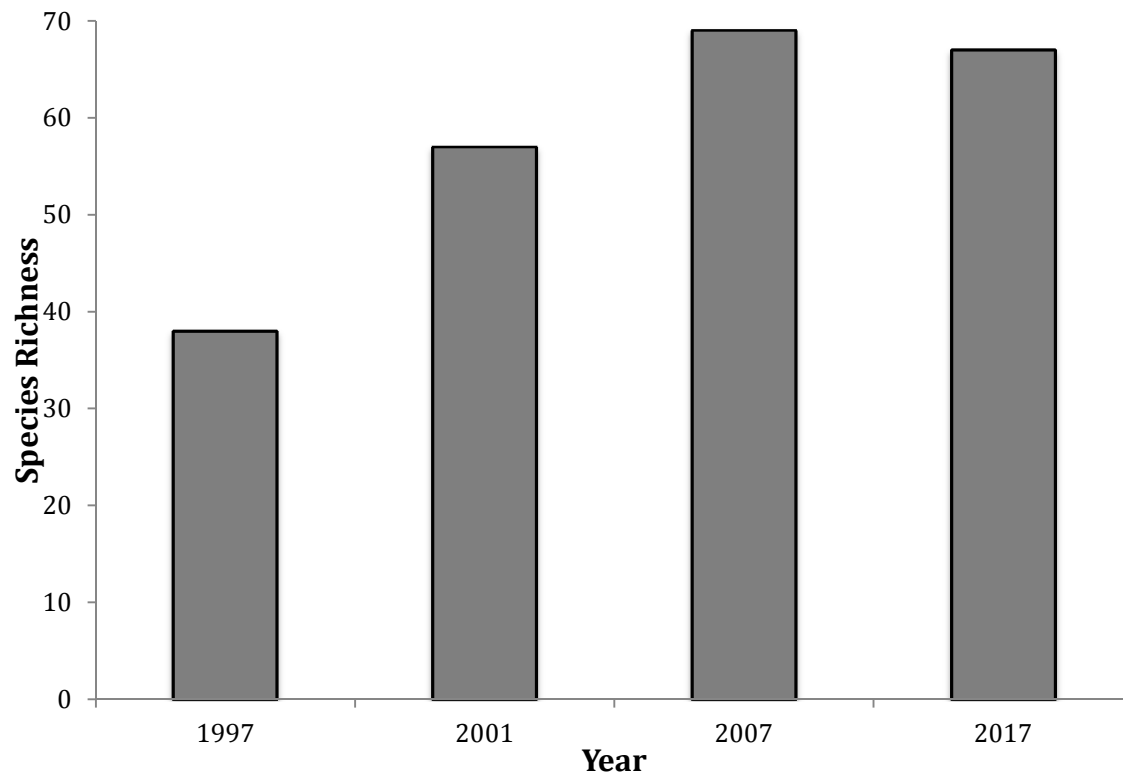


Fig. 1. Total species richness at Wizard Islet, Barkley Sound between 1997, 2001, 2007, and 2017. Species richness was found across 15 randomly selected transects at Wizard Islet's rocky intertidal zone. Graph created on Microsoft Excel version 15.36.

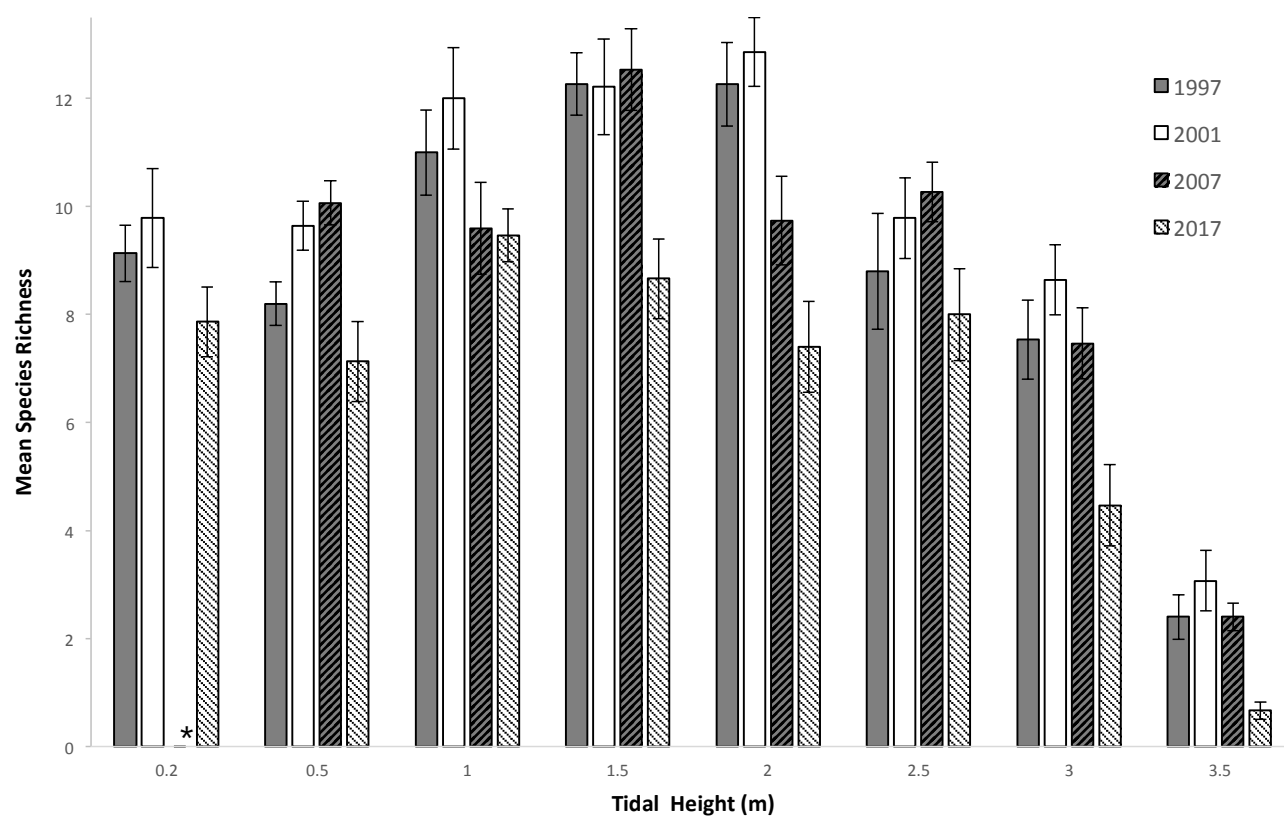


Fig. 2. Mean species richness per tidal height (m) per year at Wizard Islet's rocky intertidal zone. Each value is the result of 15 replicates. Graph created using Microsoft Excel version 15.36.
* indicates no data available

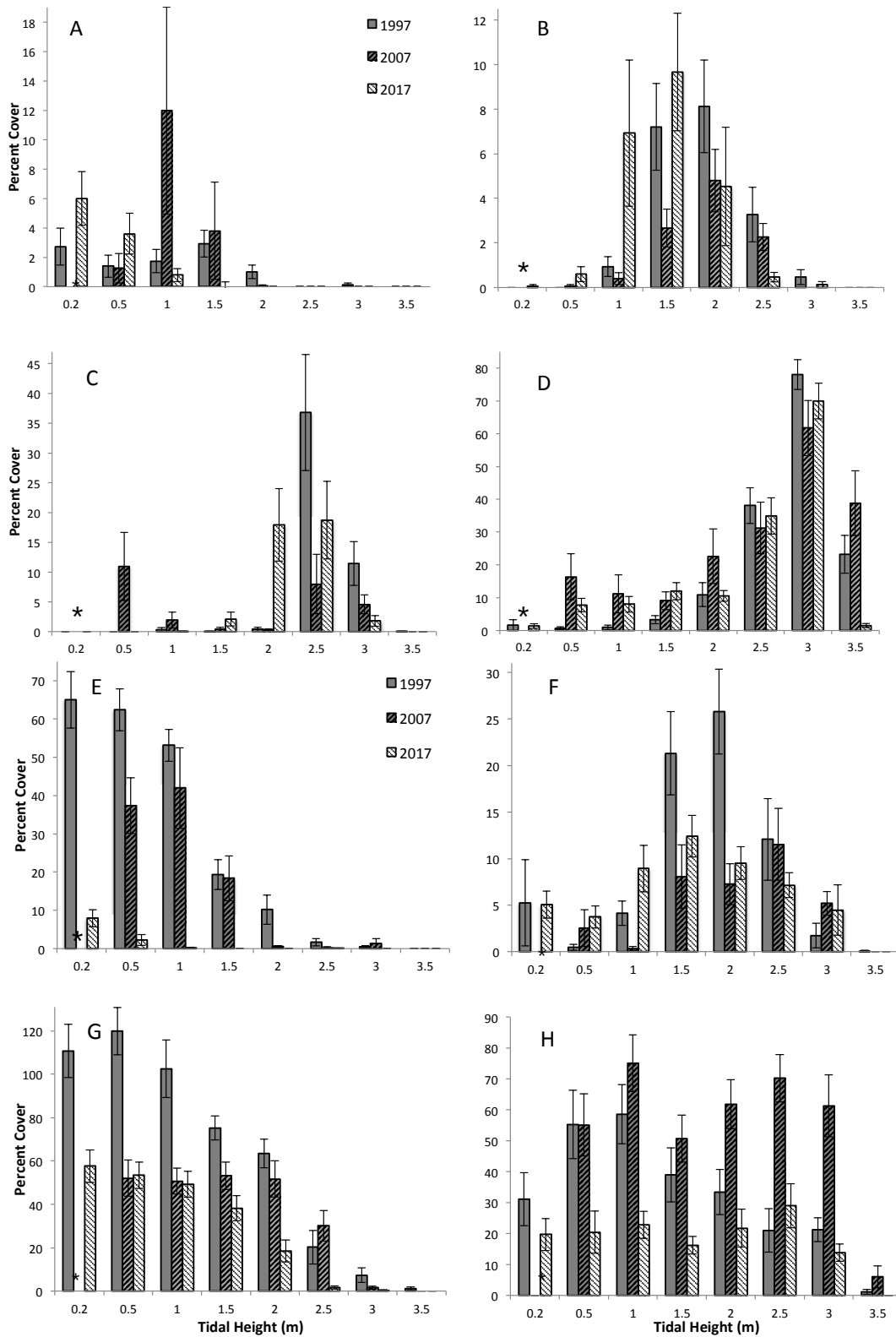


Fig. 3. Mean percent cover of the eight phyla identified at Wizard Islet's rocky intertidal zone: A) Porifera; B) Cnidaria; C) Bivalvia; D) Arthropoda; E) Bryozoa; F) Green algae; G) Rhodophyta; H) Phaeophyceae. Each value is the result of 15 replicates. Graphs were created using Microsoft Excel version 15.36. * indicates no data available

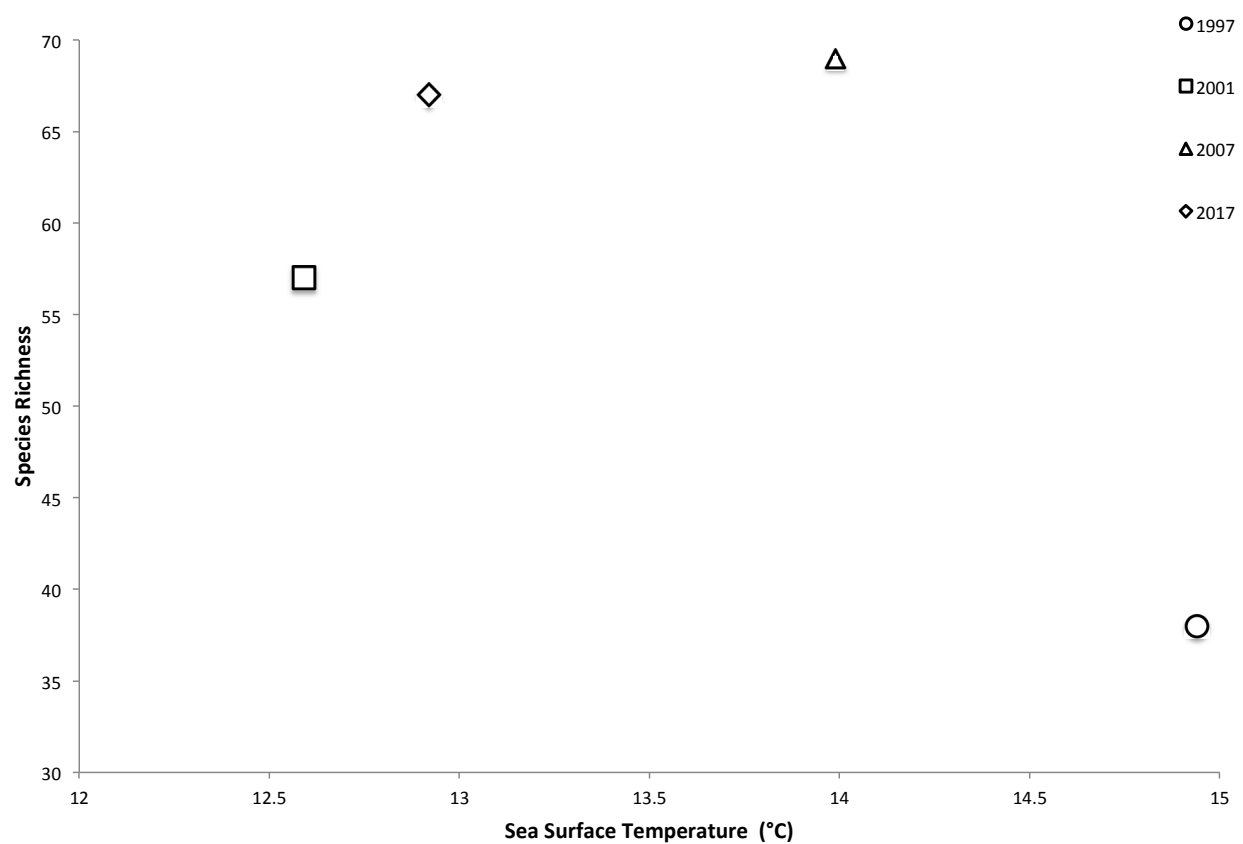


Fig. 4. The relationship between total species richness and mean sea surface temperature (°C) per year. Species richness was measured at Wizard Islet's rocky intertidal zone, Barkley Sound, whereas SST was measured at Amphitrite Lighthouse, Barkley Sound. Each species richness value is the result of 15 replicates. Graph created using Microsoft Excel version 15.36.